

EHRDyn-ICU: KDD125 lineage-consistent release candidate

Anonymous development manuscript

Five primary cohorts; large-lineage sepsis and heart failure; AF/flutter excluded at its frozen subject gate.

Known-value truth is confined to constructed environments. Retrospective EHR OPE is reliability-qualified and does not define a policy winner.

Cohorts and P/R/T definitions

P is next-state dynamics. R is a frozen reward or outcome component. T is trajectory termination or censoring, not 90-day mortality.

heart_failure: 25,694 subjects; 29,524 ICU-stay episodes (KDD121/KDD122).

sepsis: 21,837 subjects; 26,272 ICU-stay episodes (KDD121/KDD122).

aki: 16,380 subjects; 21,083 ICU-stay episodes (KDD097/KDD098R).

respiratory: 18,709 subjects; 21,679 ICU-stay episodes (KDD097/KDD098R).

shock: 37,015 subjects; 49,495 ICU-stay episodes (KDD097/KDD098R).

Superseded sepsis, AF/flutter, and heart-failure constructions are historical sensitivities only; no metric is transferred.

Reward and planning semantics

Primary sepsis R is terminal_90d_only: +100 survival, -100 death, divided by 100 for optimization and emitted once at the final valid transition.

The outcome is a hospital-discharge-origin proxy: hospital death or recorded death by discharge plus 90 days. Missing DOD without hospital death is treated as survival proxy; no explicit right-censoring field is available.

terminal_90d_plus_lactate and lactate_delta_only remain separate sensitivities and their scales are not pooled.

H1 is one-step exhaustive supported-action search. H4/H8 are four-/eight-step categorical CEM with receding-horizon first-action execution. Truncated horizons use the frozen terminal value bootstrap.

OPE atlas and claim boundary

The atlas retains 5,457 evidence-status rows: {"estimand_not_defined": 3, "not_computable_missing_probability": 54, "reported_estimator_disagreement": 27, "reported_support_fragile": 4347, "reported_supported": 1026}.

Every mathematically defined estimate is accompanied by estimand, computability, support, ESS, importance-weight tails, repeated-dataset calibration, sensitivity, and estimator disagreement.

Retrospective estimates are not true counterfactual policy values. They do not establish causal effects, treatment benefit, clinical utility, deployment readiness, or confirmatory generalization.